

SUMMARY

Software engineer building production backend and ML infrastructure at Reliance Jio. Contributed four changes upstream to PyTorch and Meta's bpfiler, with additional work in fault-tolerant distributed systems and performance tooling.

SKILLS

- **Languages** – C++, C, Python, CUDA, Java, JavaScript, TypeScript, SQL
- **Backend/Infrastructure** – Linux, Docker, Kubernetes, OpenShift, FastAPI, ClickHouse, PostgreSQL, Redis, Airflow, REST APIs, WebSockets, SSE, CI/CD
- **Systems/Performance** – CUDA kernels, multithreading, OpenMP, TLA+, warp-level primitives, shared-memory optimization
- **ML/GPU** – PyTorch, vLLM, ROCm, Hugging Face, torch.compile, NVIDIA Nsight
- **Certifications** – [Certified Kubernetes Application Developer \(CKAD\)](#), [AWS Certified Developer – Associate](#)

WORK EXPERIENCE

Reliance Jio Infocomm USA, Frisco, TX

November 2025 – Present

Software Engineer, AI

- Led backend architecture for a production 4G/5G analytics platform, building a typed-intent compiler that serves 59% of queries without an LLM, cuts p50 latency from **20s to 1s**, and reduces wrong answers from **17% to 4%**.
- Deployed a distributed backend and AI platform across a 13-node, air-gapped, **IPv6-only Kubernetes/OpenShift cluster** with AMD GPUs, integrating ClickHouse, Airflow, FastAPI, and self-hosted vLLM for production analytics and inference.
- Built an evidence-grounded RCA engine over **240 telemetry fields** and **80 RF diagnostic rules**, combining deterministic evidence with bounded multi-agent reasoning to verify causes, calibrate confidence, and abstain when evidence is insufficient.

Venture Development Center, UMass Boston, MA

September 2024 – October 2025

Software Developer

- Maintained a cloud-hosted multi-tenant production platform at **99.9% uptime**, implementing monitoring, alerting, and incident-response tooling for reliable day-to-day operations.
- Built a real-time telemetry dashboard using React and Server-Sent Events, improving application performance by **~60%** through profiling and integrating a RAG pipeline with **85% accuracy** at 15% lower cost.

Landmark Ventures, India

January 2022 – July 2023

Software Engineer Intern

- Built a React/Redux property analysis dashboard with Chart.js, enabling 3x faster deal evaluations and driving team-wide adoption.
- Cut weekly reporting from 8 hours to 1 by automating a Python pandas pipeline and training the team on Git workflows.

OPEN SOURCE CONTRIBUTIONS

- Merged 2 **PyTorch core** PRs (#158985, #173183) through maintainer review, enabling dynamic shapes for foreach ops under torch.compile and exposing rearrange in the torch.func public API.
- Merged 2 PRs into **Meta bpfiler** (#364, #369), adding IPv4 TOS and IPv6 Traffic Class matchers with BPF code generation and tests.
- Published a Donut OCR model and prescription dataset on Hugging Face, adopted by **5 community Spaces**. | [Hugging Face](#)
- Published [ChatGPT Spark](#) (4.9/5), a Chrome extension for prompt coaching, context tracking and semantic conversation search

PROJECTS

Mustard Watch Party - Distributed Video Sync | mustard.watch | [GitHub](#)

- Built a multi-instance real-time video sync engine using WebSockets, NTP-style clock correction, and predictive PI control, achieving **48 ms P95 player-measured drift at ~300 ms RTT** across browser clients.
- Designed fault-tolerant room coordination with atomic Redis Lua scripts, fencing, and idempotency keys; **model-checked retry safety** in TLA+ and preserved active rooms through instance failures.

PyTorch AutoTune - Performance Optimization Library | [PyPI](#)

- Published a training autotuner (2,000+ downloads) that picks precision, compile mode and fused optimizer by benchmarking the target model: **2.7-6.7x faster training steps than PyTorch defaults** (A100).
- Beat torch.compile alone by **2.0-6.1x** (reproduces within 1.5% when re-measured from scratch); measured why fixed configs fail: CUDA-graph mode gains 4% on ResNet-50 but 2.7x on small-batch ResNet-18.

Fused-LayerNorm-CUDA-Operator | [GitHub](#)

- Built a LayerNorm/RMSNorm CUDA kernel library for PyTorch: **fused residual-add+norm, norm-to-fp8 (E4M3) and a deterministic backward**, as drop-in modules with no graph breaks under torch.compile.
- Benchmarked RMSNorm-to-FP8 at **4.9-7.2x over eager and 1.04-1.76x over torch.compile** across A100 shapes; deterministic fwd+bwd reached 1.17-1.44x PyTorch autograd throughput at M >= 2048 (fp16).

EDUCATION

University of Massachusetts Boston, Boston, MA

September 2023 – May 2025

Master of Science in Computer Science

Lakshmi Narain College of Technology, MP, India

September 2019 – May 2023

Bachelor of Technology in Computer Science